AI Use Impact Tracker

Technical Reference & Metric Computation Guide

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This technical brief documents the calculation methods and formulas utilized across the AI Use Impact Tracker dashboard infrastructure, focusing on structural definitions, core population denominators, and data suppression rules.

1. Core Analytical Building Blocks

All metrics are calculated using individual monthly survey responses from participants. Three primary derived vectors drive the framework:

- **AI-Use Frequency Scale (0–6):** An ordinal scale capturing how frequently a participant engages with AI tools.
- **User Status:** A boolean identifier confirming if a participant uses AI at all, triggered by any score ≥ 1.
- **Impact Framework Flags:** Extracted multi-select boolean flags indicating specific directional workplace changes.

Impact profiles are segmented into two distinct directional vectors:

- **Positive Vectors:** Improved work quality or output, or creation of new income opportunities.
- **Negative Vectors:** Increased pressure to adapt, job future anxiety, operational job loss, or reduced income.
- **Neutral Safeguards:** Options noting ‘No impact’, ‘Not sure’, or ‘Other’ are classified as neutral and do not contribute to directional balances.

2. Core Population Indicators

These base descriptive metrics are calculated independently for every demographic stratum and month block.

Total Respondent Count (N)

The straightforward absolute headcount representing all individuals who finalized the survey inside a designated cohort.

Adoption Rate

The proportion of the surveyed population that actively employs AI tools.

$$Adoption\ \ Rate = \frac{1}{N} \sum User\_Status$$

Average Frequency Score

The mean frequency score across all respondents who provided a valid non-null frequency answer (N′).

$$Average\ \ Frequency = \frac{1}{N'} \sum Frequency\_Score$$

3. The Impact Denominator Framework

To isolate real trends, workspace impact metrics do not use the full population headcount (N) as their base. Instead, they rely strictly on a verified sub-population called the Impact Denominator (N_impact).

To be included in this specific denominator, a respondent must fulfill two rules simultaneously:

- **1. Active Engagement:** They must be a confirmed AI user (Frequency score ≥ 1).
- **2. Valid Response:** They must have provided an explicit answer to the impact question, including choosing baseline answers like ‘No impact’ or ‘Not sure’.

> *Methodological Purpose: Non-users are removed from impact calculations because a participant with zero regular interaction cannot report direct workspace shifts. This step ensures that the resulting percentages accurately reflect the experiences of the user base without downward dilution.*

4. Composite Strategic Summary Indices

The framework synthesizes individual flags into two big-picture composite metrics built off the Impact Denominator.

Net Impact Index (NII)

This index tracks the broad directional leaning of the cohort by subtracting the negative impact share from the positive impact share.

$$NII = Positive\_Share - Negative\_Share$$

Weighted Impact Index (WII)

The primary headline parameter of the tracker dashboard. Rather than checking raw directional splits, it applies standardized analytical severity weights to individual answers to map economic severity:

Workplace Impact Choice Flag	Assigned Severity Weight
Created new job or income opportunities	+1.00
Improved my work quality or output	+0.50

Made me worry about the future of my job or industry	-0.25
Increased pressure to adapt or work faster	-0.50
Reduced my income or made it harder to find work	-0.75
Caused me to lose my job	-1.00
No impact / Another impact / Not sure	0.00

To compute the WII, the system calculates a total personal score for each respondent based on their combined flags, then extracts an arithmetic mean over the Impact Denominator:

$$WII = \frac{1}{N} \sum Personal\_Score$$

5. Governance Rules & Controls

- **Suppression Boundary:** Any demographic matrix cell tracking fewer than 50 total respondents is automatically suppressed to ensure confidentiality and filter out raw statistical noise.
- **Multi-Month Pooling:** Time-window filters (such as rolling 3- or 6-month selectors) aggregate data using sample-size weighted means across constituent cells to ensure mathematical balance in multi-month views.
- **Dose-Response Layout:** The dose-response grid tracks the Net Impact Index across isolated usage groups to evaluate behavioral trends without introducing claims of true direct causality.